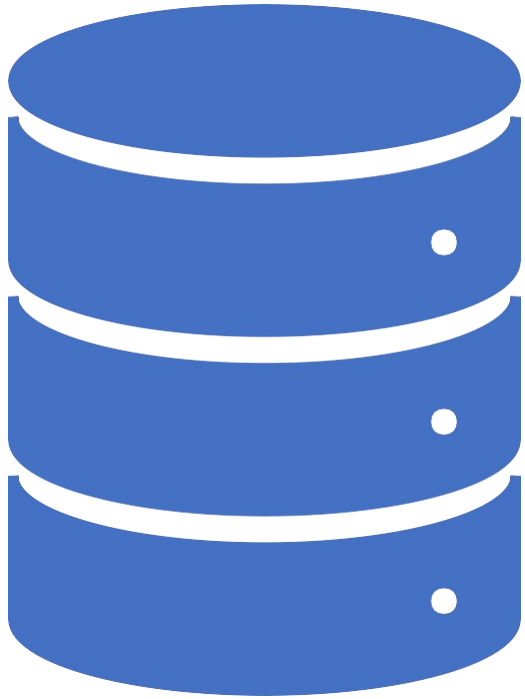
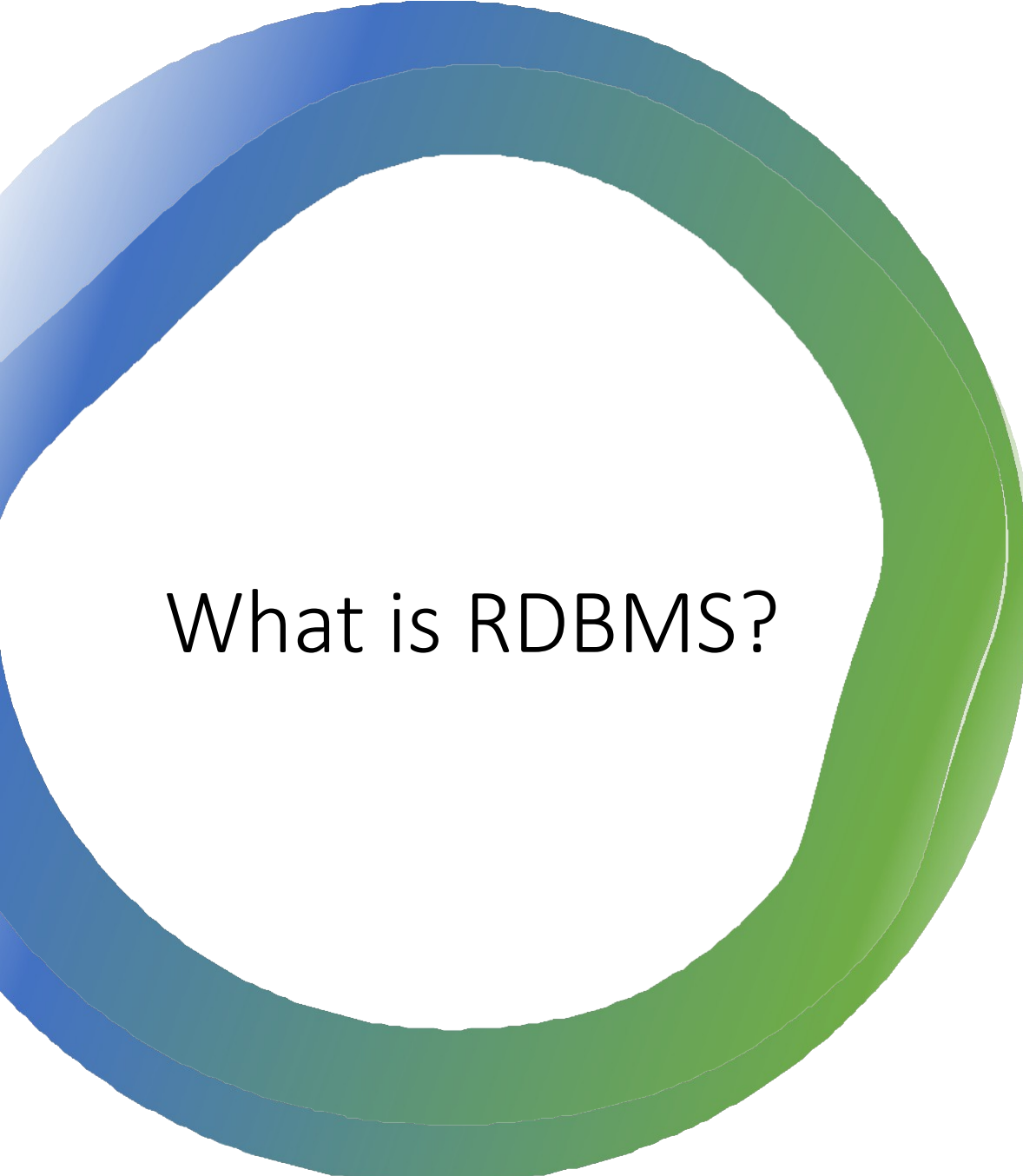




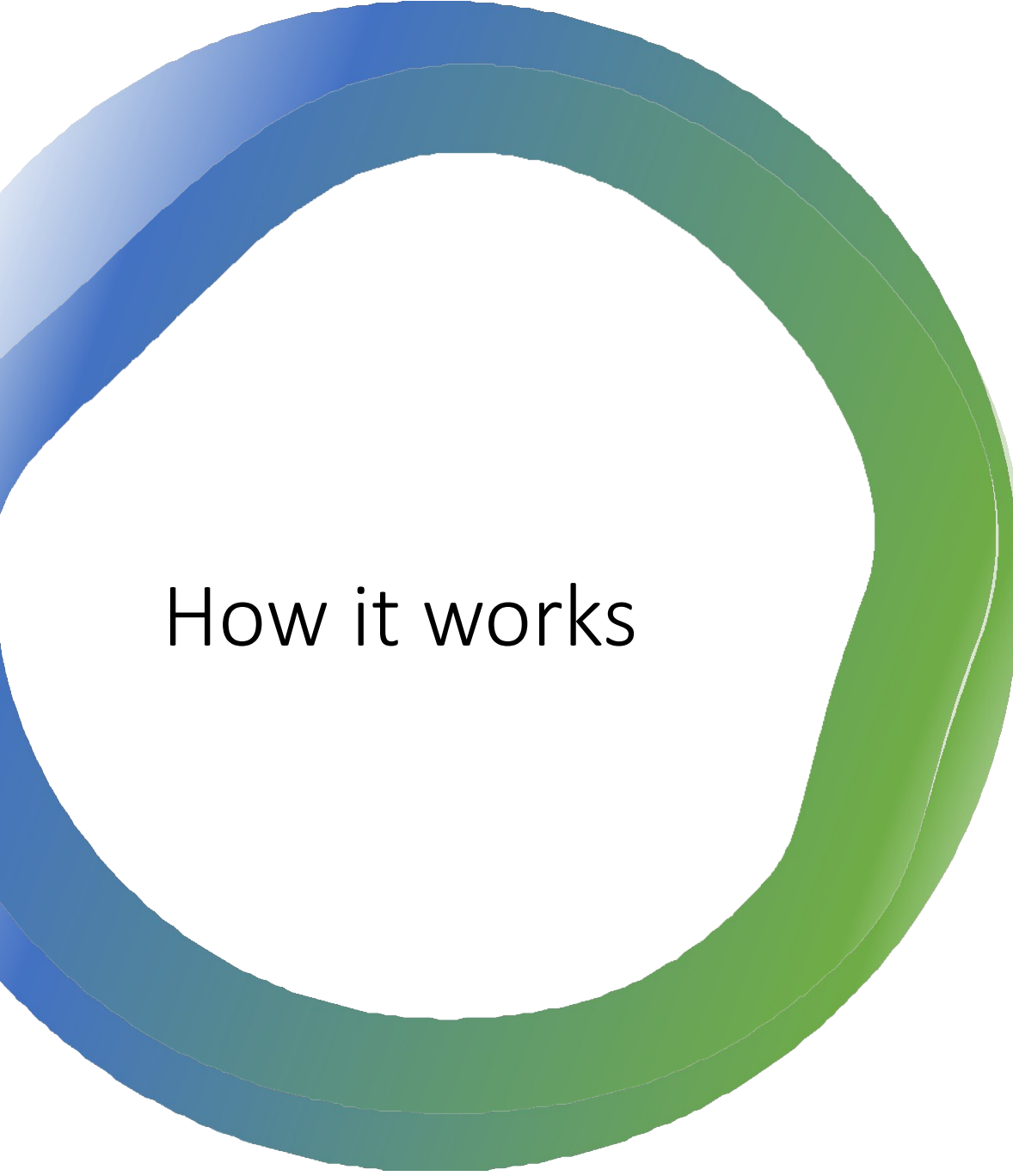
Database Systems

[*What is RDBMS?*]



What is RDBMS?

- RDBMS stands for ***Relational Database Management System***.
- All modern database management systems like SQL, MS SQL Server, IBM DB2, ORACLE, My-SQL, and Microsoft Access are based on RDBMS.
- It is called Relational Database Management System (RDBMS) because it is based on the relational model introduced by E.F. Codd.



How it works

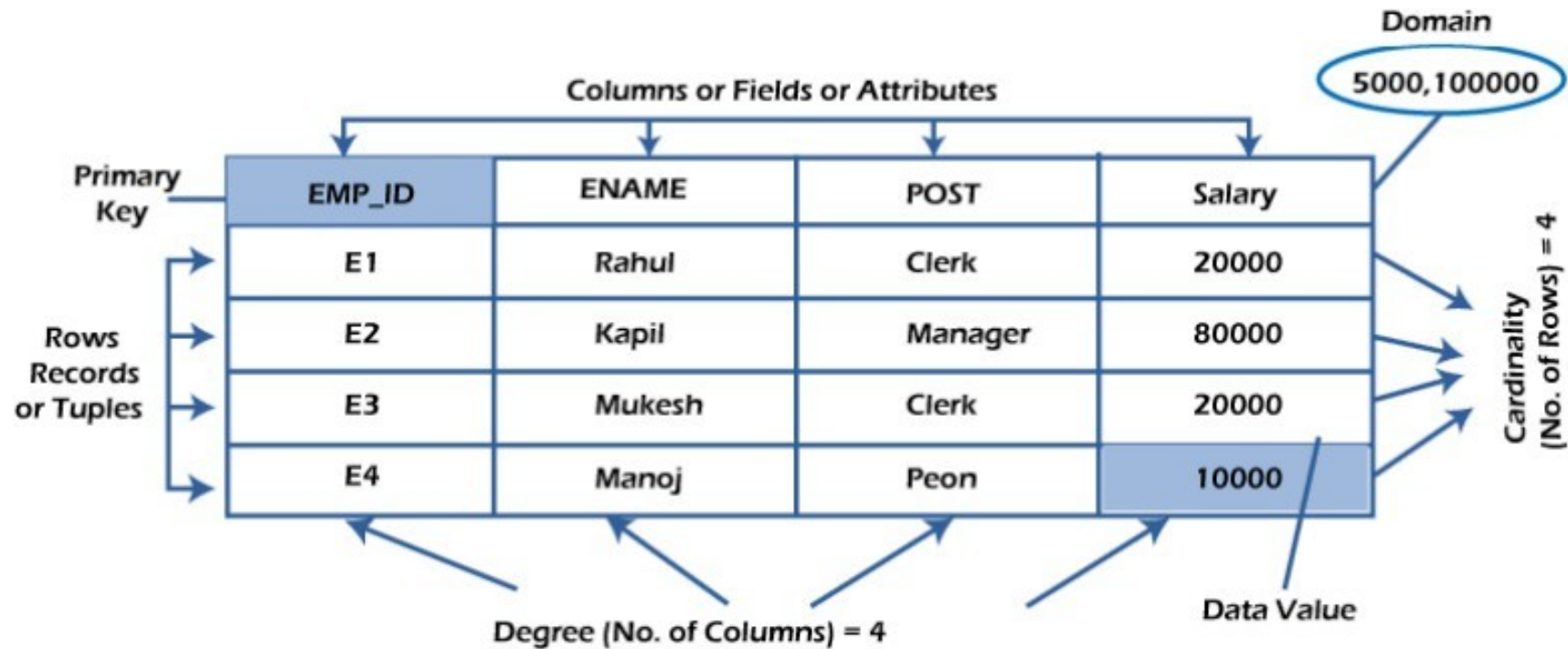
- Data is represented in terms of tuples (rows) in RDBMS.
- A relational database is the most commonly used database. It contains several tables, and each table has its primary key.
- Due to a collection of an organized set of tables, data can be accessed easily in RDBMS.

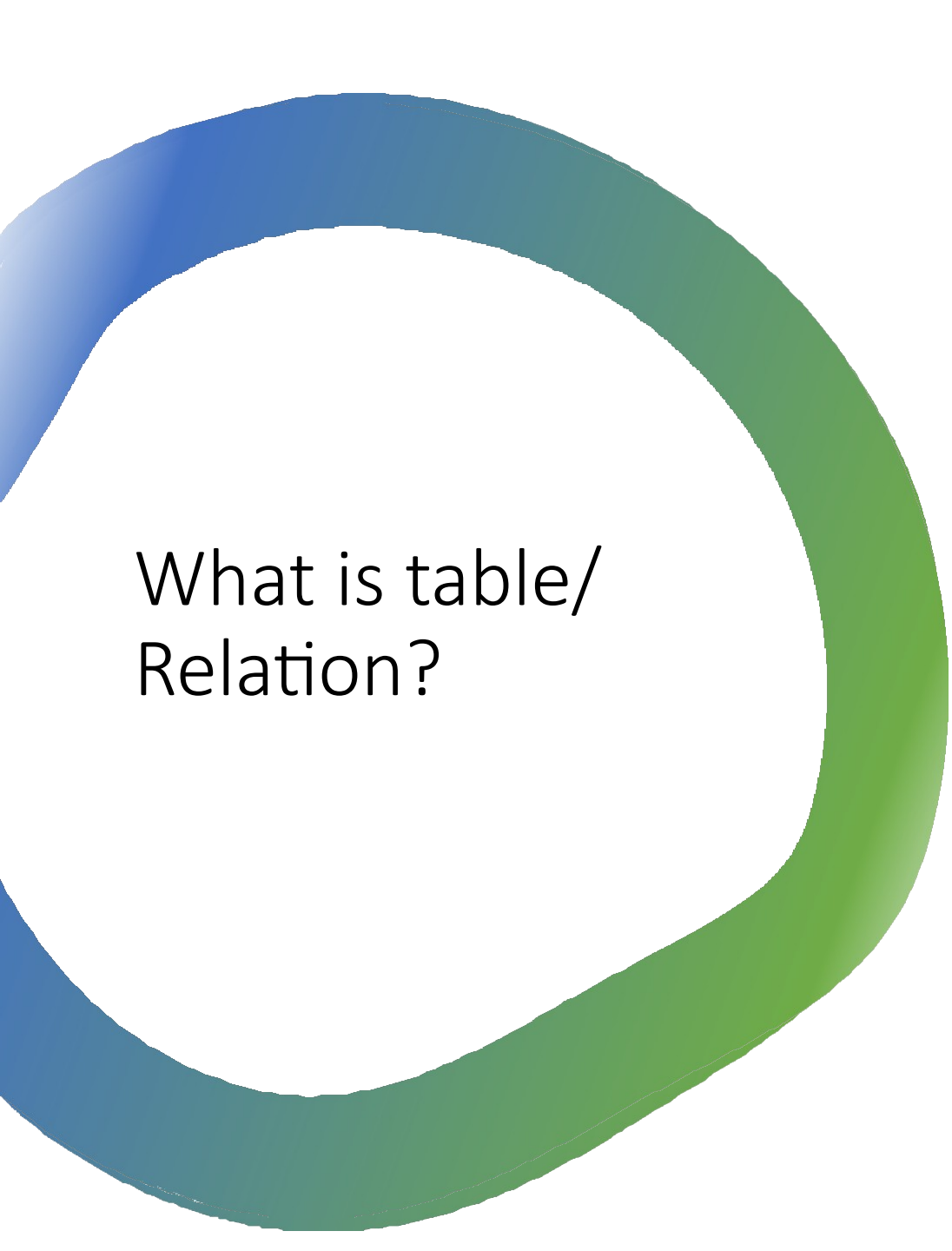


Brief History of RDBMS

- From 1970 to 1972, E.F. Codd published a paper to propose using a relational database model.
- RDBMS is originally based on E.F. Codd's relational model invention.

Terminologies of RDBMS





What is table/ Relation?

- Everything in a relational database is stored in the form of relations.
- The RDBMS database uses tables to store data.
- A **table** is a collection of related data entries and contains rows and columns to store data.
- Each table represents some real-world objects such as person, place, or event about which information is collected.
- The organized collection of data into a relational table is known as the **logical view** of the database.



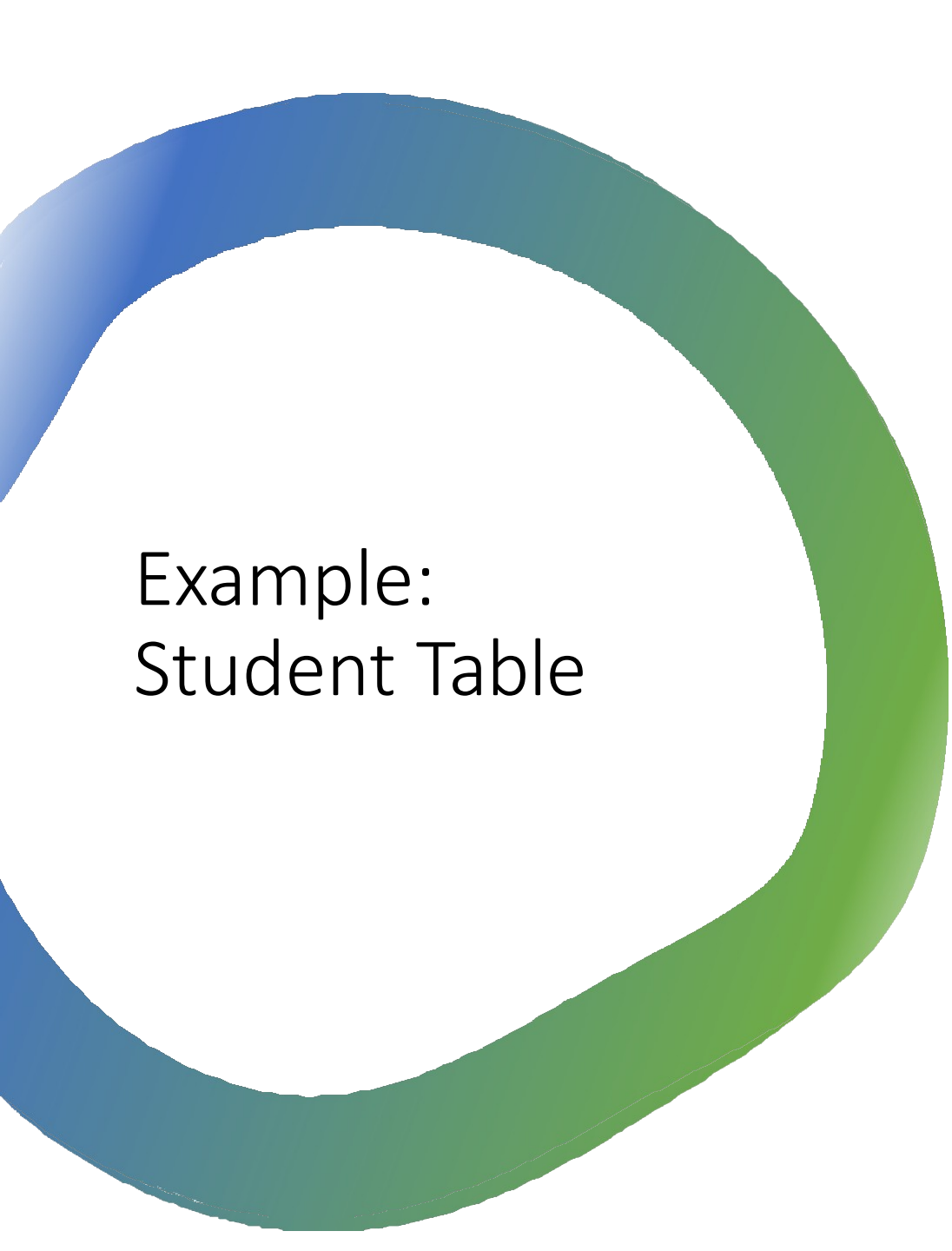
Properties of a Relation

Each relation has a ***unique name*** by which it is identified in the database.

Relation ***does not*** contain ***duplicate*** tuples.

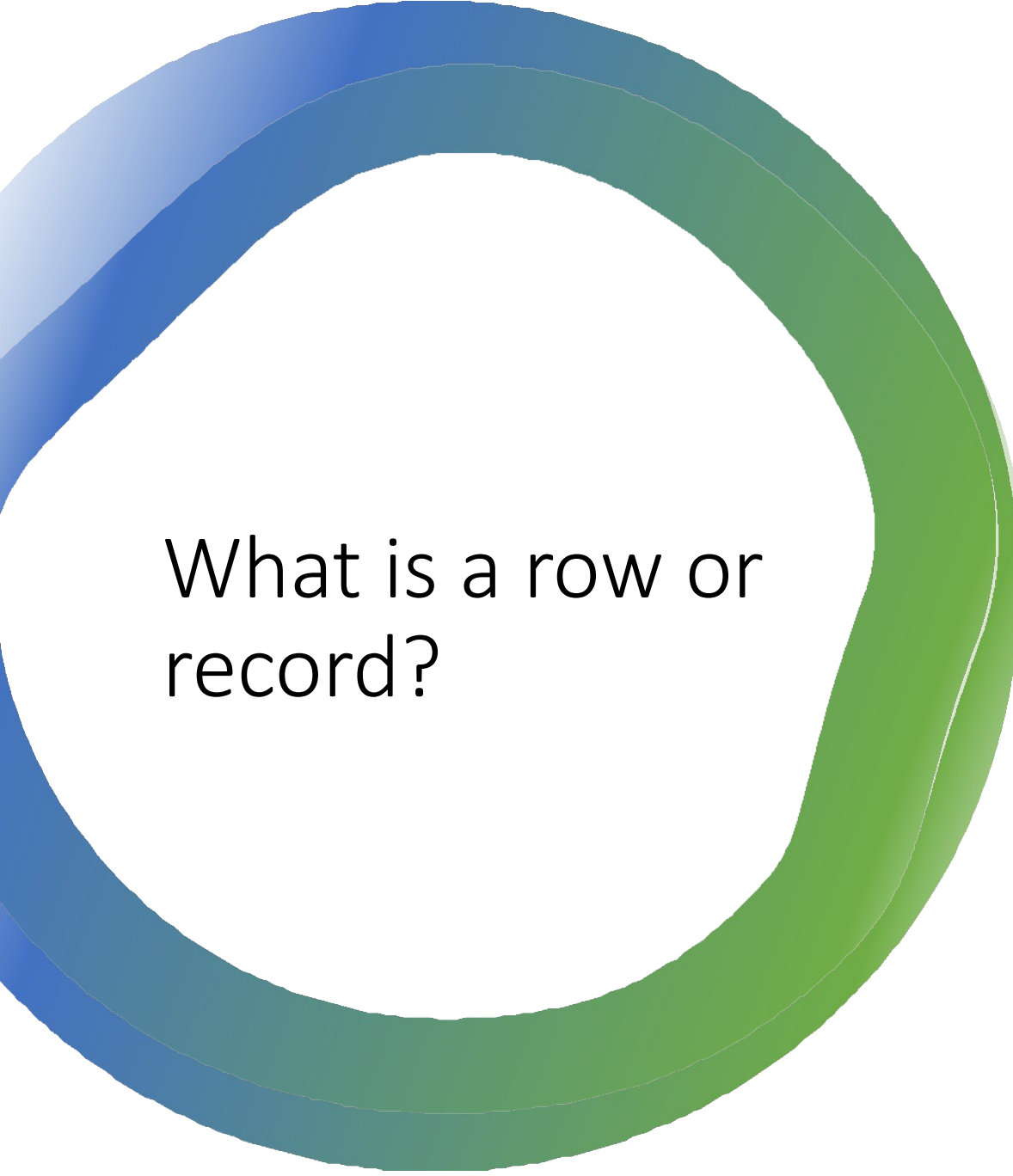
The tuples of a relation have ***no specific order***.

All attributes in a relation are ***atomic***, i.e., each cell of a relation contains exactly one value.



Example:
Student Table

ID	NAME	AGE	COURSE
1	Ayyaz	24	B. Tech
2	Kamil	20	CA
3	Saqib	21	BCA
4	Kamran	22	MCA
5	Saeed	26	BSC



What is a row or record?

- A **row** of a table is also called a **record** or **tuple**.
- It contains the specific information of each entry in the table.
- It is a horizontal entity in the table.
- For example, The above table contains 5 records.

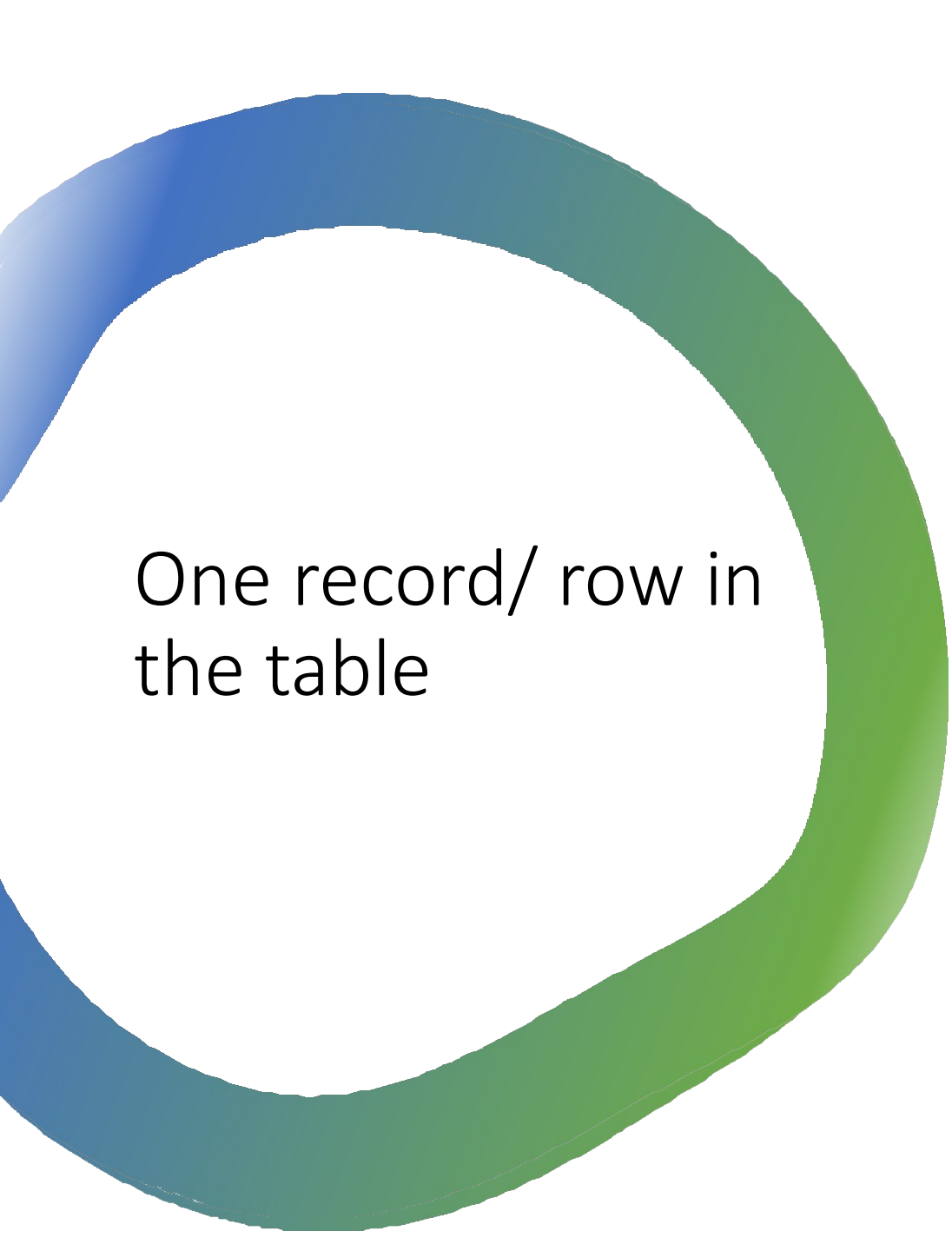


Properties of a Row

No two tuples are identical to each other in all their entries.

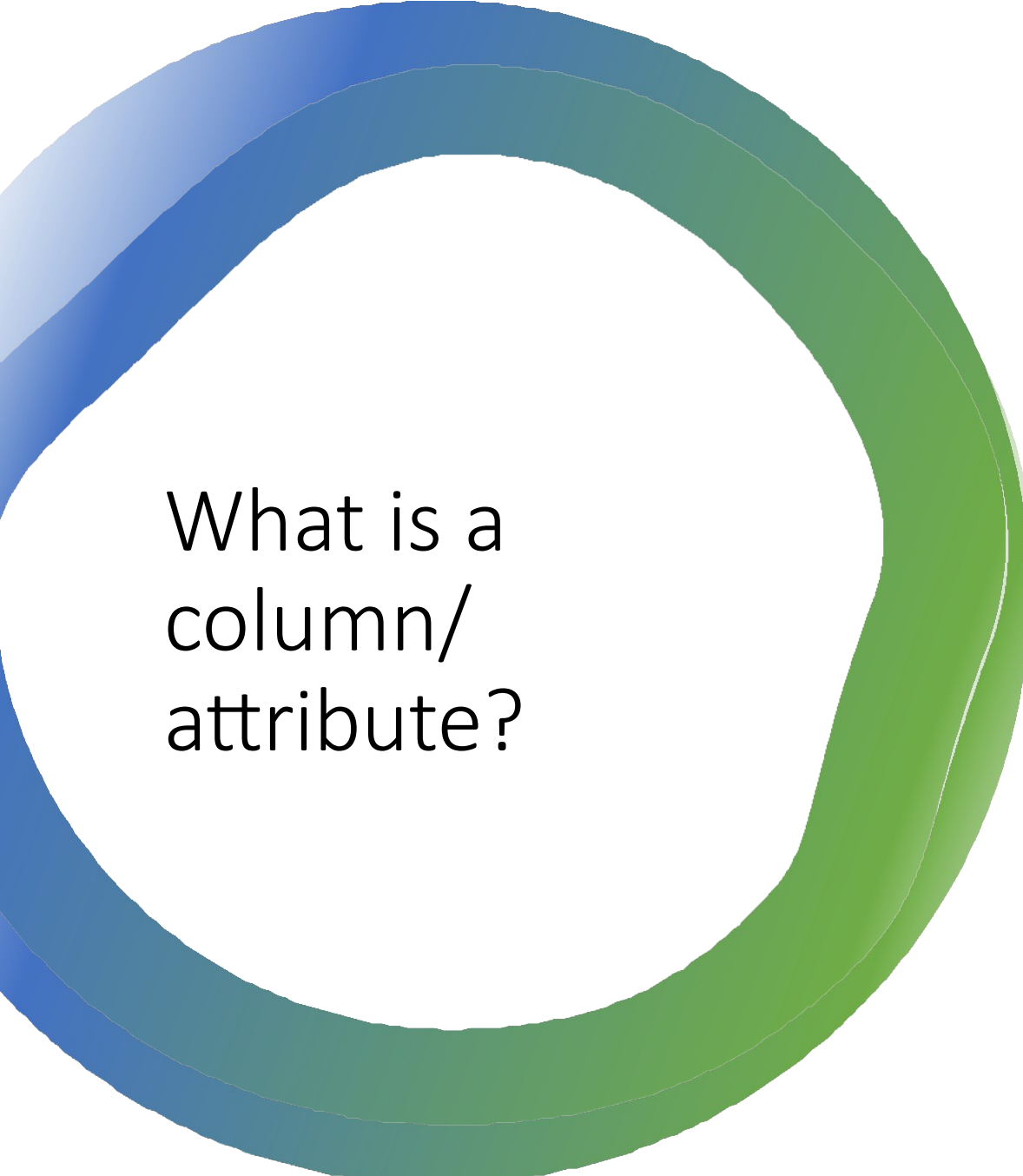
All tuples of the relation have the ***same format*** and the ***same number of entries***.

The order of the tuple is ***irrelevant***. They are ***identified by their content***, not by their position.



One record/ row in
the table

ID	NAME	AGE	COURSE
1	Ayyaz	24	B. Tech
ID	NAME	AGE	COURSE
1	Ayyaz	24	B. Tech
2	Kamil	20	CA
3	Saqib	21	BCA
4	Kamran	22	MCA
5	Saeed	26	BSC



What is a
column/
attribute?

- A ***column*** is a vertical entity in the table which contains all information associated with a specific field in a table.
- For example, "**name**" is a column in the above table which contains all information about a student's name.



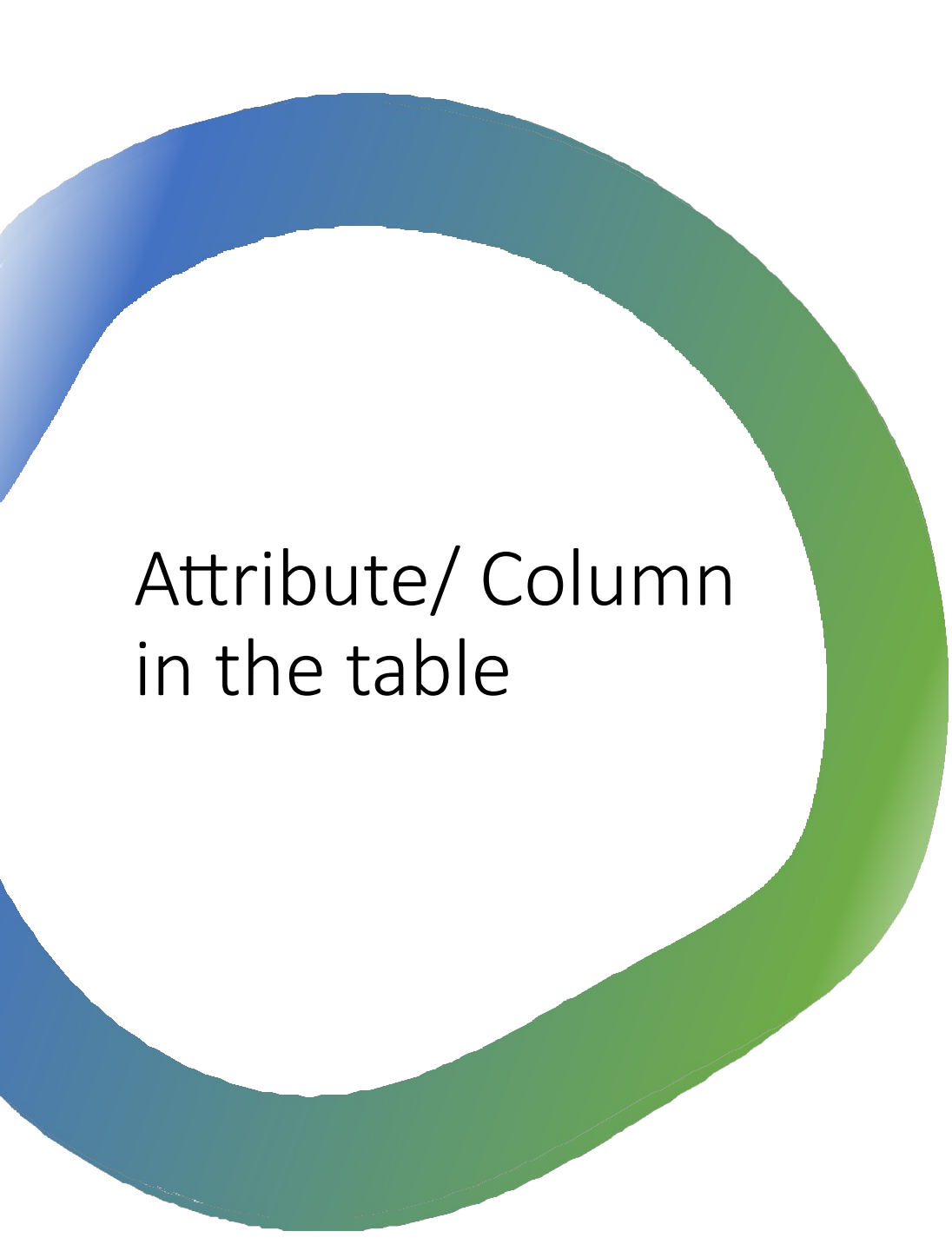
Properties of an Attribute

Every attribute of a relation ***must have a name.***

Null values are permitted for the attributes.

Default values can be specified for an attribute ***automatically inserted*** if no other value is specified for an attribute.

Attributes that ***uniquely identify each tuple*** of a relation are the **primary key**.



Attribute/ Column
in the table

ID	NAME	AGE	COURSE
1	Ayyaz	24	B. Tech
2	Kamil	20	CA
3	Saqib	21	BCA
4	Kamran	22	MCA
5	Saeed	26	BSC

NAME

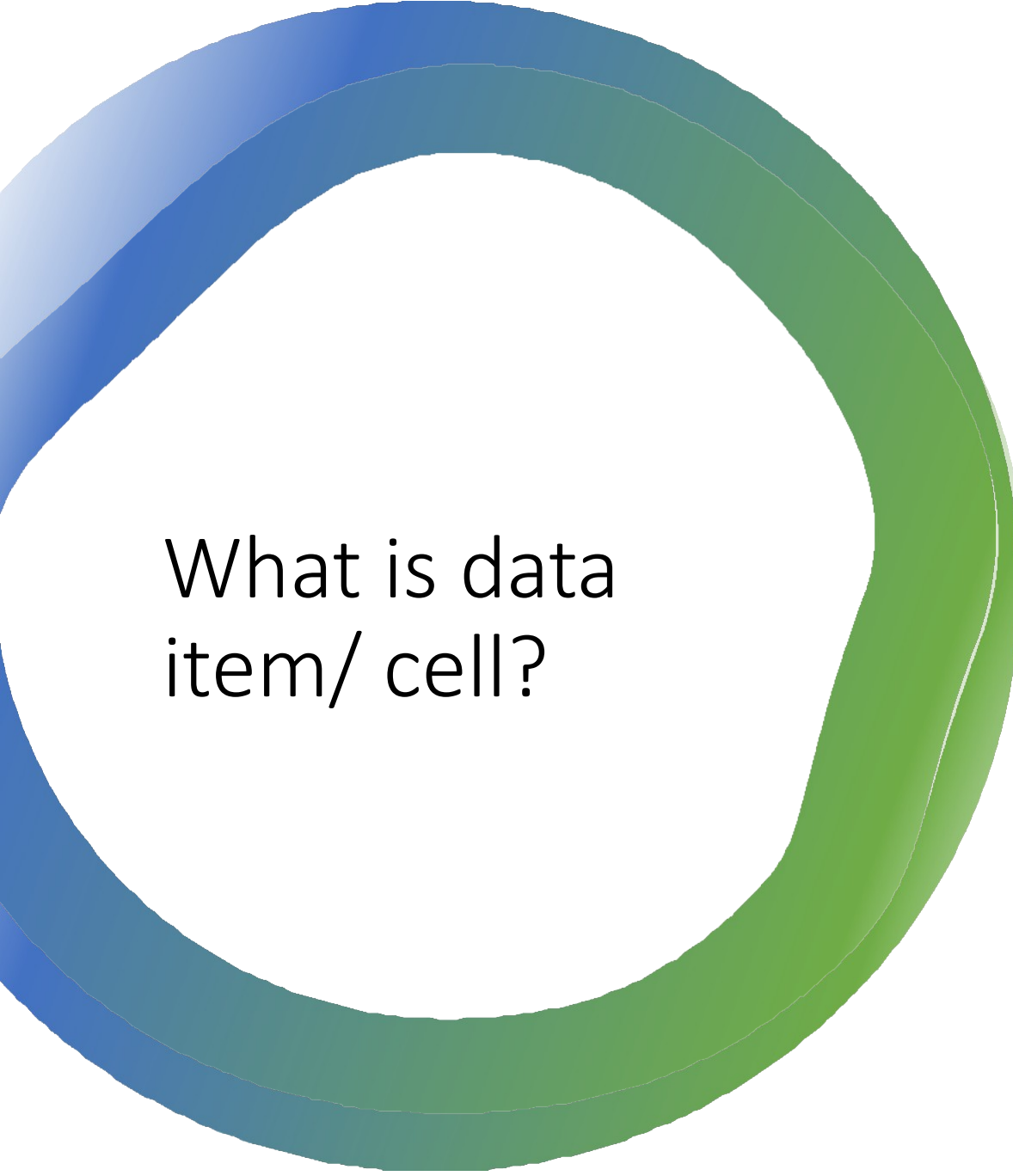
Ayyaz

Kamil

Saqib

Kamran

Saeed



What is data
item/ cell?

- The ***smallest unit of data*** in the table is the individual data item.
- It is stored at the *intersection of tuples and attributes*.



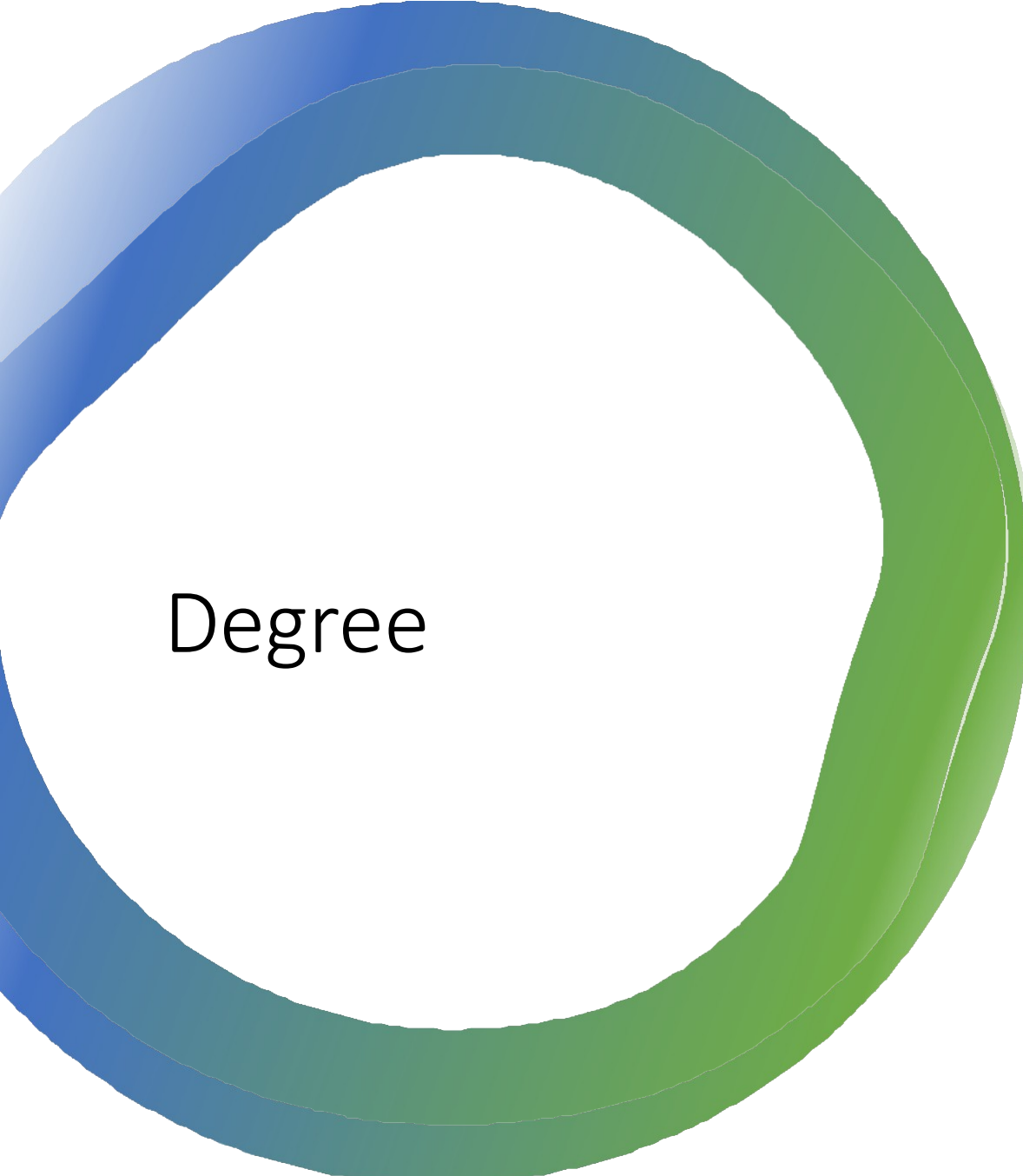
Properties of Data Items

Data items are ***atomic***.

The data items for an attribute ***should be drawn from the same domain***.

In the below example, the data item in the student table consists of Ajeet, 24 and Btech, etc

ID	NAME	AGE	COURSE
1	Ayyaz	24	B. Tech

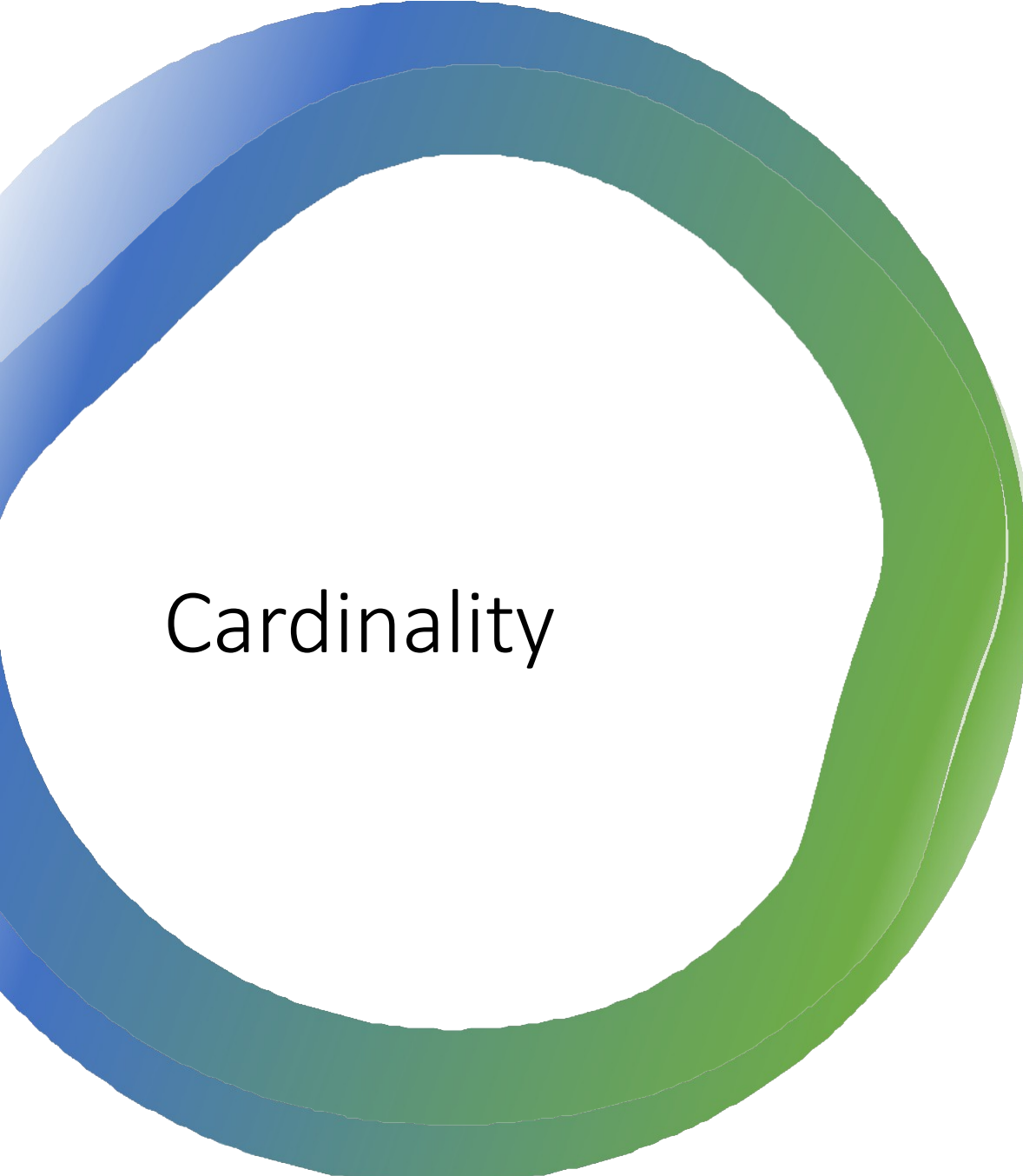


Degree

- The ***total number of attributes*** that comprise a relation is known as the **degree** of the table.

ID	NAME	AGE	COURSE
1	Ayyaz	24	B. Tech
2	Kamil	20	CA
3	Saqib	21	BCA
4	Kamran	22	MCA
5	Saeed	26	BSC

The student table has 4 attributes, and its ***degree is 4.***

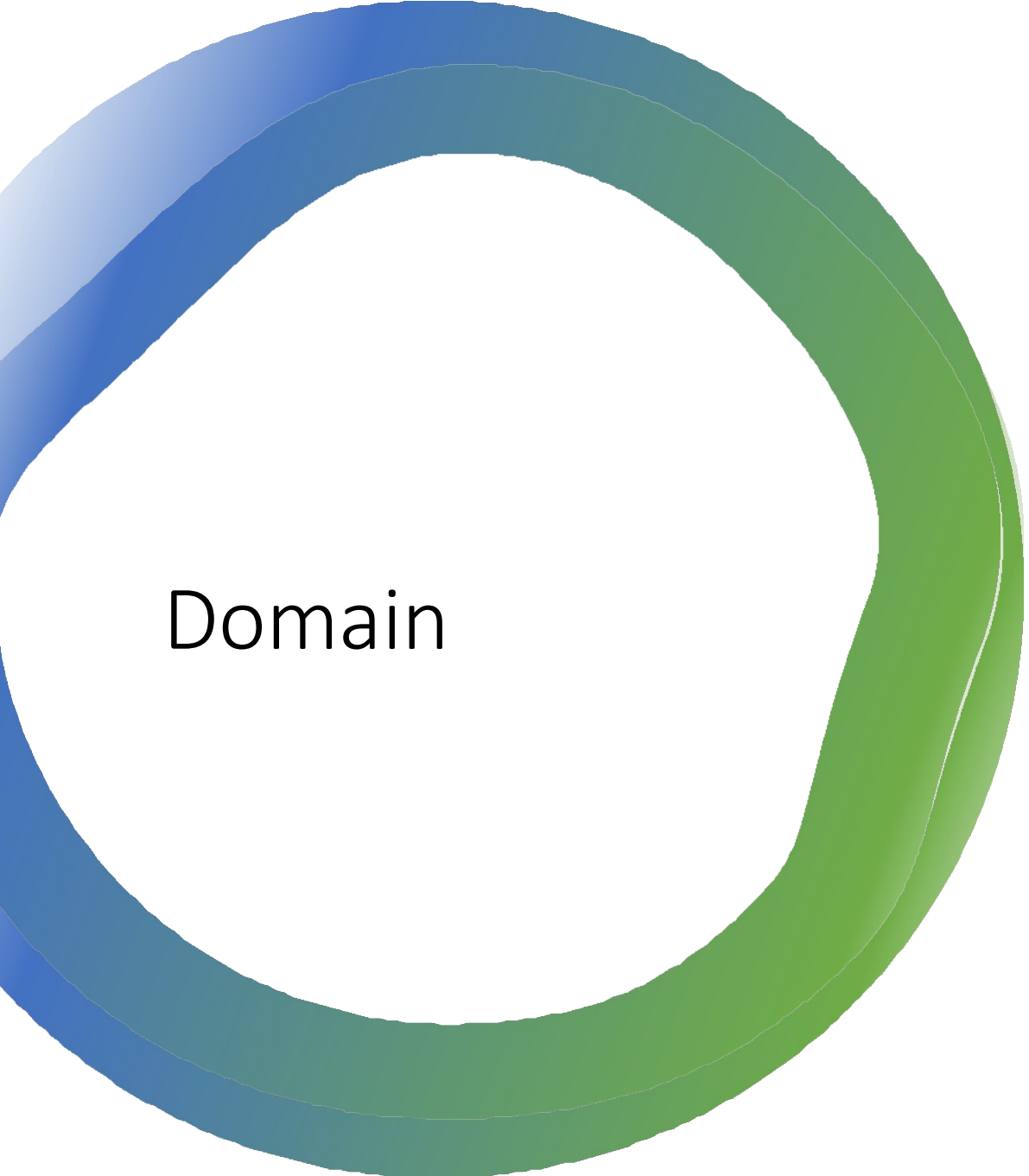


Cardinality

- The total number of tuples at any one time in a relation is known as the table's cardinality.
- The relation whose cardinality is 0 is called an empty table.

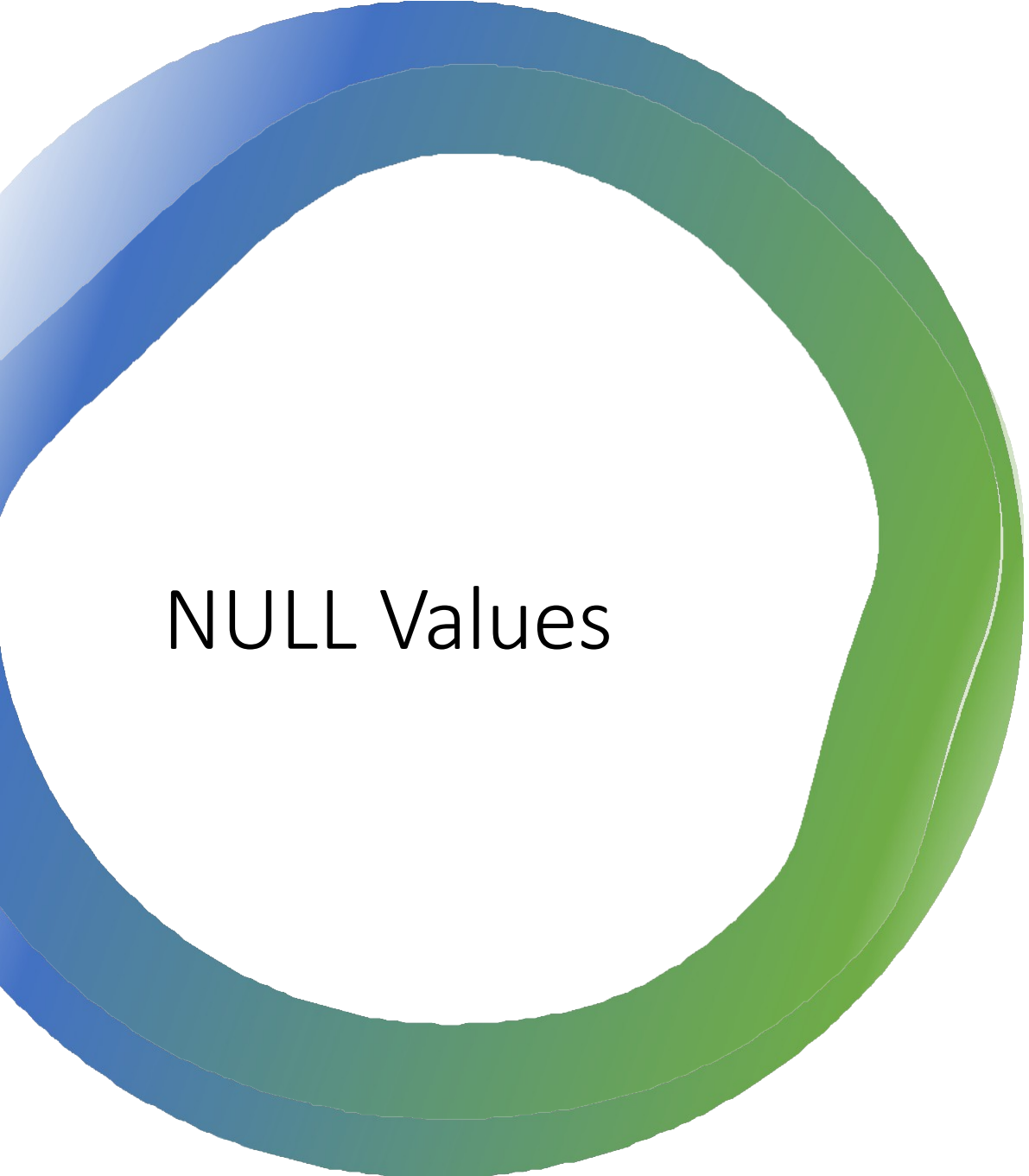
ID	NAME	AGE	COURSE
1	Ayyaz	24	B. Tech
2	Kamil	20	CA
3	Saqib	21	BCA
4	Kamran	22	MCA
5	Saeed	26	BSC

The student table has 5 rows, and its cardinality is 5.



Domain

- The **domain** refers to the *possible values each attribute can contain*.
- It can be specified using standard data types such as integers, floating numbers, etc.
- For example,
 - An attribute entitled **Marital_Status** may be limited to *married* or *unmarried* values.



NULL Values

- The **NULL value** of the table specifies that the field has been *left blank* during record creation.
- It is different from the value filled with zero or a field that contains space.



Data Integrity

Following are the categories of data integrity exist with each RDBMS:

Entity integrity:

- It specifies that there should be **no duplicate rows** in a table.

Domain integrity:

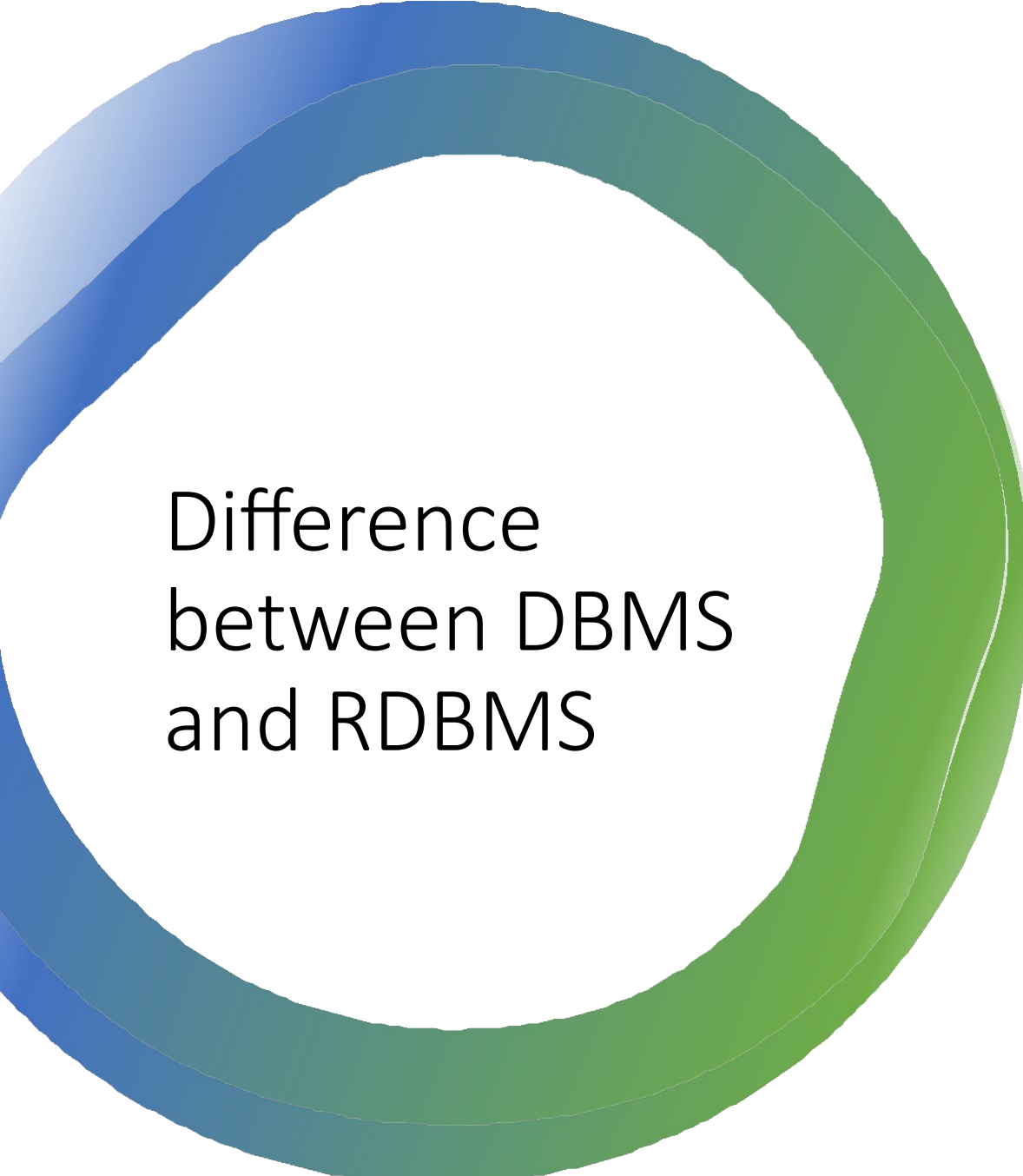
- It enforces **valid entries** for a given column by *restricting the type, the format, or the range of values.*

Referential integrity:

- Specifies that **rows cannot be deleted**, which are used

User-defined integrity:

- It **enforces some specific business rules** defined by users. These rules are different from the entity, domain, or referential integrity.



Difference between DBMS and RDBMS

- Although DBMS and RDBMS both are used to store information in physical database but there are some remarkable differences between them.
- The main differences between DBMS and RDBMS are given below:

DBMS	RDBMS
DBMS applications store data as file .	RDBMS applications store data in a tabular form .
In DBMS, data is generally stored in either a hierarchical form or a navigational form.	In RDBMS, the tables have an identifier called primary key and the data values are stored in the form of tables.
Normalization is not present in DBMS.	Normalization is present in RDBMS.
DBMS does not apply any security with regards to data manipulation.	RDBMS defines the integrity constraint for the purpose of ACID (Atomicity, Consistency, Isolation and Durability) property.
DBMS uses file system to store data, so there will be no relation between the tables .	in RDBMS, data values are stored in the form of tables, so a relationship between these data values will be stored in the form of a table as well.
DBMS has to provide some uniform methods to access the stored information.	RDBMS system supports a tabular structure of the data and a relationship between them to access the stored information.
DBMS does not support distributed database .	RDBMS supports distributed database .
DBMS is meant to be for small organization and deal with small data . it supports single user .	RDBMS is designed to handle large amount of data . it supports multiple users .
Examples of DBMS are file systems, xml etc.	Example of RDBMS are mysql, postgre, sql server, oracle etc.